RoomCleaner

A cable-driven robot that picks laundry off the floor

Jasper Buntinx · Technical presentation · Oct 27, 2026



02

The winch station had to survive its own motor

Heat, torque and a spool that kept walking off its shaft

CALCULATED

An aluminium fin pulls heat out of a sealed printed shell



TRIGGER

Sealing the station in PETG made the motor mount its hottest, most loaded joint

OPTIONS

Seven fixes weighed: vents, a cooler driver, ASA, a thermistor, a metal fin …

CHOICE

A 1/8 in 6061 fin, narrowed to 44 × 250 mm to fit the station

CALCULATED

Thicker spreads heat better, but 1/8 in was the knee

1/16 in

54 %

Fin efficiency

Lightest and easiest to cut, but gives up almost half of its fin effect.

1/8 in

CHOSEN

68 %

Fin efficiency

Most of the gain at half the mass of 1/4 in. The stock I sheared and drilled.

1/4 in

81 %

Fin efficiency

Best spreading, but twice the aluminium hanging on a printed wall mount.

CALCULATED

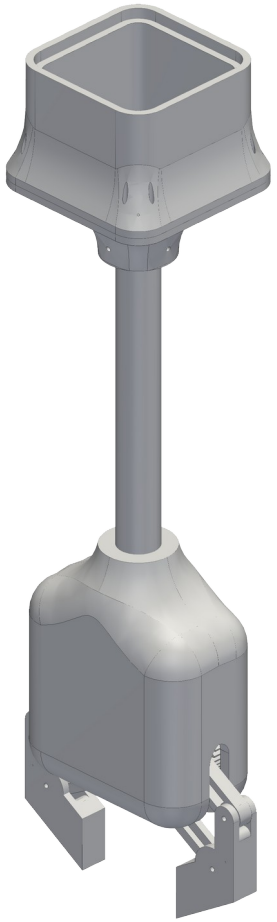
My first full gripper was over four times its mass budget

2.06 kg

upper-bound mass of gripper v6 against a 450 g budget

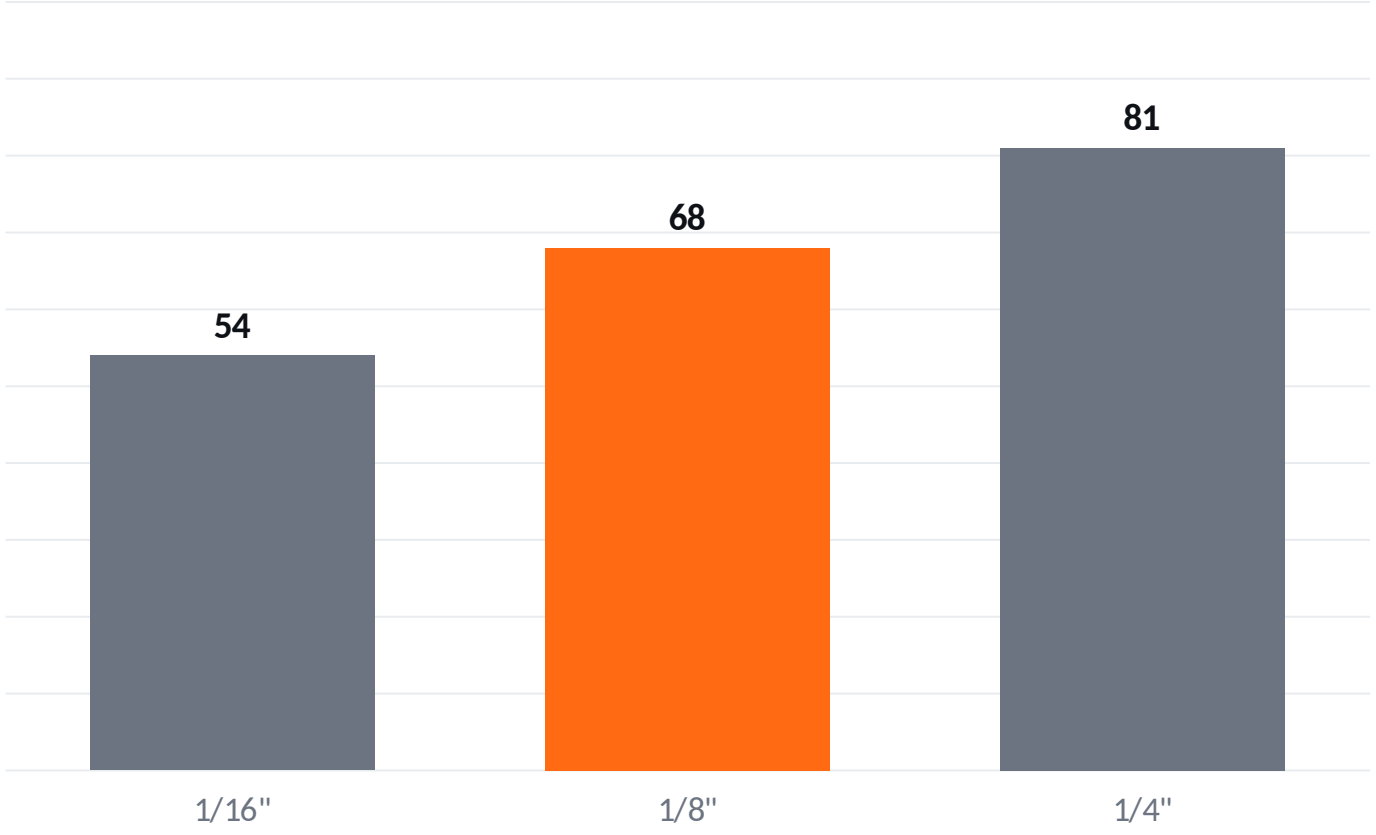
$$m = \Sigma\, \rho \cdot V \;\cdot\; \rho(\mathrm{PETG}) = 1.27\ \mathrm{g/cm^3}, 100\ \%\ \text{infill bound}$$

Source: CAD volume of every body in the v6 STEP, solid PETG



CALCULATED

Doubling the plate past 1/8 in buys only 13 more points



Fin efficiency η = tanh(mL)/mL with m = √(2h/kt).

1/8 in gets 68 % for half the aluminium of 1/4 in.

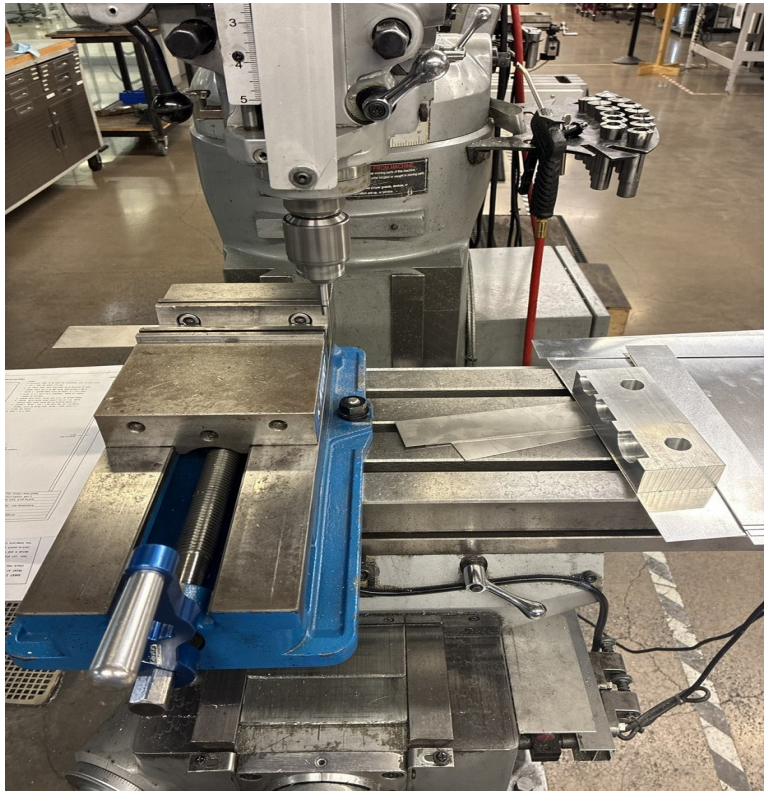
That mass sits on a printed wall mount, so the knee wins.

MEASURED

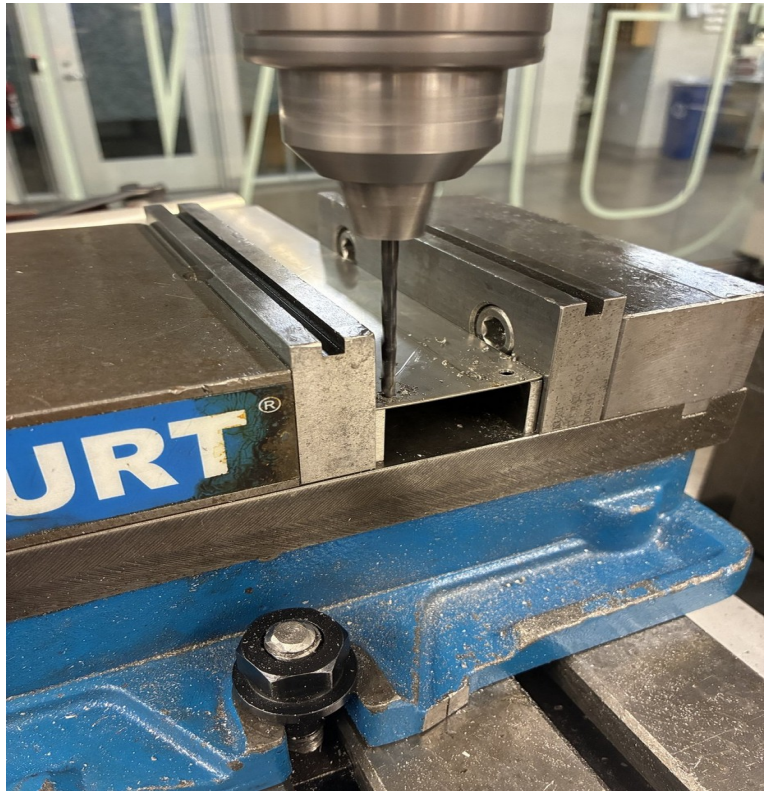
I sheared, laid out and drilled the fins myself



1 Sheared blanks against the Letter drawing



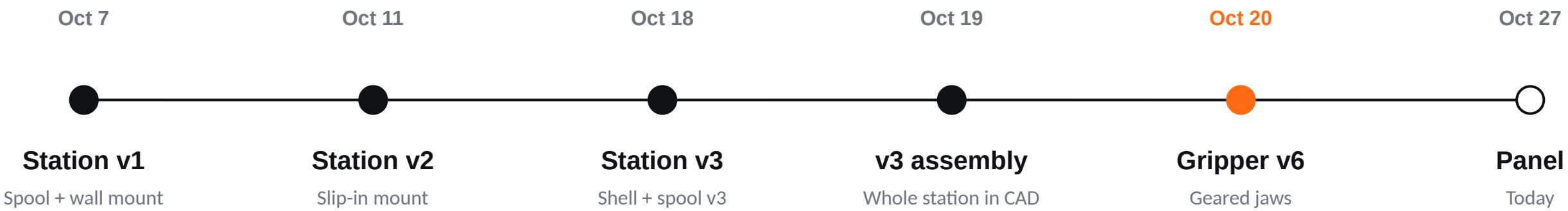
2 Mill set up from the inch drawing



3 Drilling the Ø3.4 mounting holes

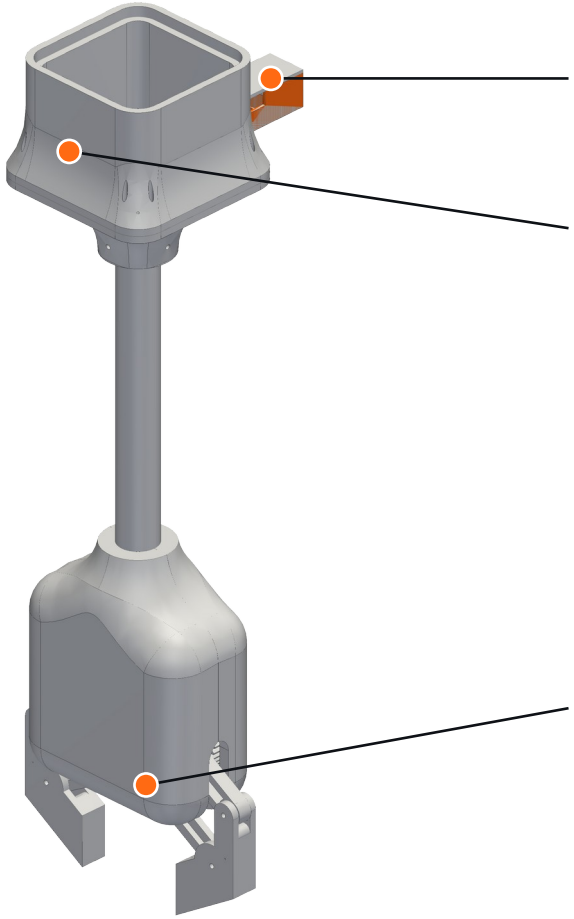
MEASURED

Five design snapshots in two weeks



SIMULATED

The camera rides on the narrow side so it can see the jaws' approach



Lens 512 mm above the floor, looking straight down.

12 % of the view is blocked by the claw itself.

Lines can only touch the bay above 1.98 m at ≥ 30° yaw — so tasks are capped at 1.6 m.

WHAT I'D TELL MYSELF ON DAY ONE

Three things this robot taught me

01

Measure before you trust a margin

Sample lesson text — one line, plain words.

02

Every interface is a decision

Sample lesson text — one line, plain words.

03

Make failure visible early

Sample lesson text — one line, plain words.

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Palette, type and evidence tags

Ink
0E1116 · Text, dark slides

Paper
FFFFFF · Content background

Fog
E9ECEF · Cards, mats

Steel
6B7480 · Secondary text

Signal
FF6A13 · The one thing to see

Tint
FFE6D6 · Chosen card only

Claim title — Arial Bold 32

EYEBROW — ARIAL BOLD 11, TRACKED

Body — Calibri 16. Left aligned, ≤ 30 words a slide, units on every number.

Caption / source — Calibri Italic 11

MEASURED

CALCULATED

SIMULATED

PLANNED